Rohan Malhotra

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EDUCATION

New York University, Courant Institute of Mathematical Sciences — New York, NY
B.A. in Mathematics and Computer Science (Double Major) — *Expected May 2027*

- **GPA:** 3.7/4.0 | Accelerated three-year degree | **Graduate course:** Mathematical Techniques for CS Applications
- **Coursework:** Data Structures, Algorithms, Operating Systems, Computer Organization, Software Engineering

EXPERIENCE

IBM — Dallas, TX
Robotics Engineer Intern — *Jun 2026 – Aug 2026*

- Built a real-time pipeline connecting YOLO11 inference to the Boston Dynamics Spot SDK via OpenCV and gRPC.
- Designed a multithreaded service with lock-free inference queues; demoed live retrieval at IBM DevCon.
- Open-sourced weights, training pipelines, and docs; IBM sales and client-success teams adopted the demo.

Applications Developer Intern, IBM Consulting — *May 2026 – Aug 2026*

- Built an on-prem EPM assistant that turns plain-language requests into reviewable JSON using RAG and MCP tools.
- Connected banking APIs to ERP and EPM via Oracle Integration Cloud and migrated Excel workflows to Oracle Cloud.

Kalshi — New York, NY
Quantitative Developer — *Jan 2026 – May 2026*

- Built a C++ hedging engine that converts contract prices (cents, probability, bps) into real-time exposure.
- Built a Next.js/Python engine that recommends hedge size and compares cost vs. tail risk using FRED and BLS data.
- Wrote C++ risk tools, containerized the runtime with Docker, and deployed it to Render.

Hume Center for National Security and Technology, Virginia Tech — Blacksburg, VA
Aerospace Researcher — *Nov 2024 – May 2025*

- Programmed imaging and signal-processing tests in C for ContentCube, a CubeSat deployed from the ISS in 2025.
- Engineered a test environment simulating concurrent imaging workloads to resolve race conditions and memory bugs.

Aress Software — New York, NY
Machine Learning Intern — *Jun 2025 – Aug 2025*

- Trained and deployed PyTorch models on **500K+ multimodal records** across Azure and AWS.
- Designed NoSQL pipelines unifying 10+ data sources into one analytics layer; optimized SQL queries for speed.

PROJECTS

NightShift | *FastAPI, Playwright, Next.js, Stripe* — 2026

- Built a platform that finds jobs, fills applications via browser automation, and verifies confirmation emails.
- Added scheduled runs, Stripe subscriptions, and a live dashboard with a full application log.

EPM Wizard | *FastAPI, React, Groovy, Playwright* — 2026

- Built a tool that turns plain-English requests into Oracle EPM forms and rules, with preview and approval steps.

Refrax | refrax.io | *C++, Next.js, TypeScript, PostgreSQL* — Jun 2025 – Feb 2026

- Designed low-latency REST/WebSocket pipelines streaming market data into a PostgreSQL research environment.
- Implemented C++ risk engines (Greeks, P&L simulation, portfolio exposure) optimized for parallel execution.

LEADERSHIP

PIVOT (AMP Lab) | vtpivot.org — Blacksburg, VA
Founder & Project Lead — *Feb 2024 – Dec 2025*

- Founded a multi-university research group and built its C++ signal engine with automated Raspberry Pi execution.

SKILLS

Languages: Python, C++, C, Java, TypeScript, JavaScript, SQL, Rust, Groovy
Frameworks: React, Next.js, FastAPI, Flask, Node.js, Playwright, PyTorch, OpenCV
Infrastructure: Linux, Git, Docker, Kubernetes, AWS, Azure, PostgreSQL, gRPC, WebSockets, REST APIs